



LAB TASK 3

Question 1:

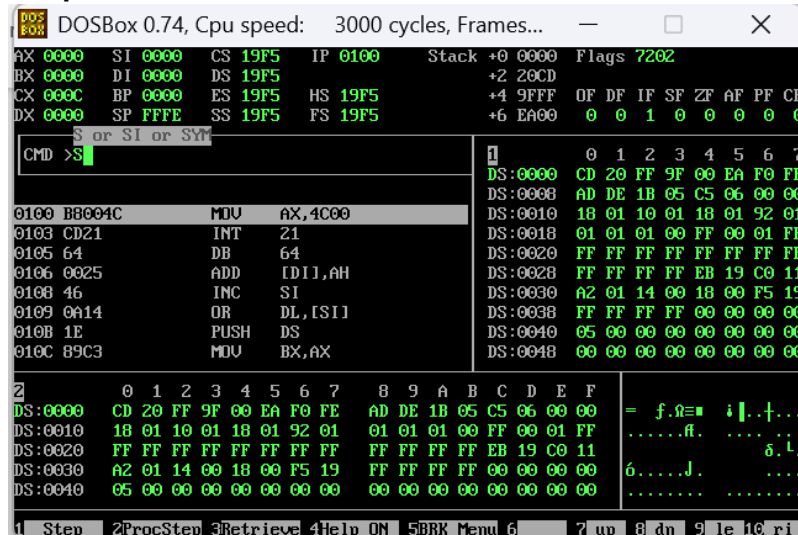
CODE

```
1.asm
1  [org 0x0100]
2
3  mov ax, 0x4c00
4  int 0x21
5
6  score: db 100
7  population: dw 2500h
8  initial: db 'F'
9  numbers: db 10, 20, 30, 40
```

EXPLANATION

- ✓ **score: db 100** -> stores value of 100(decimal value) in define byte 'score'. This will occupy **1byte** with the value of 0064h.
- ✓ **population: dw 2500h**: Stores the value 9472(decimal value) in define word 'population'. This will occupy **2byte** with the value of 2500h.
- ✓ **initial: db 'F'**: stores the value 70(ASCII for 'F' in decimal) in define word variable 'initial'. This will occupy **1 byte** with the value 0046h.
- ✓ **numbers: db 10,20,30,40**: Stores the values in define byte variable.

Output



MEMORY USAGE TABLE

VARIABLE	Type	Size (byte)	Address	Value
SCORE	Define byte	1	0100	0064h / 100d
POPULATION	Define word	2	0101-0102	2500h / 9472d
INITIAL	Define byte	1	0103	0046h / 79d
NUMBER[0]	Define byte	1	0104	000Ah / 10
NUMBER[1]	Define byte	1	0105	0014h / 20
NUMBER[2]	Define byte	1	0106	001Eh / 30
NUMBER[3]	Define byte	1	0107	0028h / 40

Question 2:

Code

```
[org 0x0100]

mov bl, [score]
mov cx, [population]
mov dh, [initial]
mov [population], ax
mov [score], cl

mov ax, 0x4c00
int 0x21

score: db 100
population: dw 2500h
initial: db 'F'
numbers: db 10, 20, 30, 40
```

OUTPUT

DOSBox 0.74, Cpu speed: 3000 cycles, Frames...									
AX 0000	SI 0000	CS 19F5	IP 0113	Stack +0 0000	Flags 7200				
BX 0064	DI 0000	DS 19F5		+2 20CD					
CX 2500	BP 0000	ES 19F5	HS 19F5	+4 9FFF	OF DF IF SF ZF AF PF CF				
DX 4600	SP FFFE	SS 19F5	FS 19F5	+6 EA00	0 0 1 0 0 0 0 0				
CMD >SS									
010F 8B0E1B01	MOV	[011B],CL	DS:0000	CD 20 FF 9F 00 EA F0 FE	0 1 2 3 4 5 6 7				
0113 8B004C	MOV	AX,4C00	DS:0008	AD DE 1B 05 C5 06 00 00					
0116 CD21	INT	21	DS:0010	18 01 10 01 18 01 92 01					
0118 0000	ADD	[BX+SI],AL	DS:0018	01 01 01 00 FF 00 01 FF					
011A 00460A	ADD	[BP+0A],AL	DS:0020	FF FF FF FF FF FF FF FF					
011D 141E	ADC	AL,1E	DS:0028	FF FF FF FF EB 19 C0 11					
011F 2BF6	SUB	DH,DH	DS:0030	A2 01 14 00 18 00 F5 19					
0121 0000	ADD	[BX+SI],AL	DS:0038	FF FF FF FF 00 00 00 00					
0123 8B46F6	MOV	AX,[BP-0A]	DS:0040	05 00 00 00 00 00 00 00					
DS:0000	CD 20 FF 9F 00 EA F0 FE	AD DE 1B 05 C5 06 00 00							
DS:0010	18 01 10 01 18 01 92 01	01 01 01 00 FF 00 01 FF							
DS:0020	FF FF FF FF FF FF FF FF	FF FF FF FF EB 19 C0 11							
DS:0030	A2 01 14 00 18 00 F5 19	FF FF FF FF 00 00 00 00							
DS:0040	05 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00							

EXPLANATION

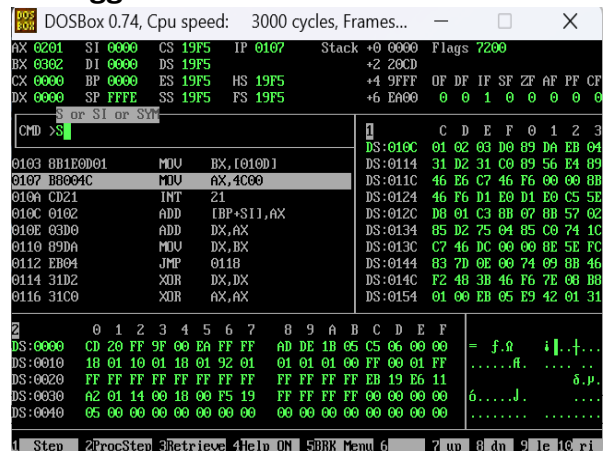
- score value [0064] is stored in bl register, population value [2500] stored in cx register and initial value [0046] stored in dh register.
- The value of ax is moved in 'population'
- The value stored in cl register is moved in score variable.

Question 3:

Code

```
SM 3.asm
1  [org 0x0100]
2
3  mov ax, [score]
4  mov bx, [score + 1]
5  mov ax, 0x4c00
6  int 0x21
7
8  score: db 1, 2, 3
9
```

Debugger



Explanation of the problem

- We are using **db** and it only store 1 byte number. But when we are moving the value of store [0] in the register **ax** which can store 16-bit number. Therefore, there is the memory mismatch.
- In the output showing on the right, **0201** is stored in **ax** register instead of **0001**.

Possible solution

- To fix this memory mismatch we have two solutions
- Either use **define word** instead of **define byte**.
- Or use **al/ah** register instead of **ax**

Question 4:

CODE

```
[org 0x0100]
mov [var1], 99h
mov ax, [var2]
mov bl, var1
mov ax, 0x4c00
int 0x21
var1 dw 1234h
var2 db 12h
```

1. **Mov [var1], 99h:** program doesn't know whether 99h is 8-bit or 16-bit.
2. **Mov ax, [var2]:** var2 is define byte (8-bit)number and we are moving it to ax which will store 16-bit value, **use ah/al instead of ax.**
3. **mov bl, var1:** we are moving address to bl not value, also var1 stores 16-bit number and bl can store upto 8 bit only